

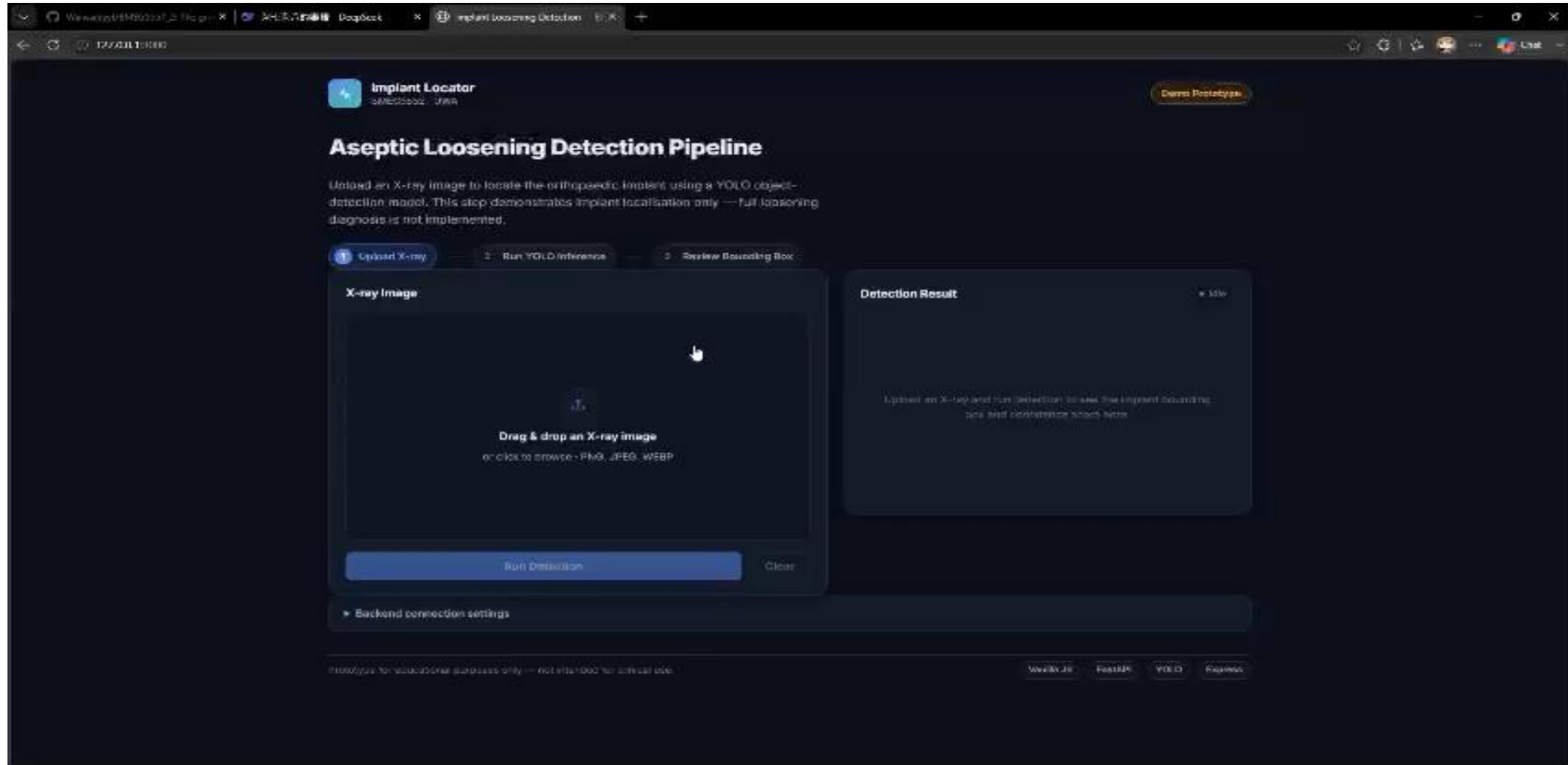
# Concept Development Presentation

Team 4

# Concept Description (10%)

- Include:
  - Video of our prototype (mentioning whether the video is analysing images that the AI was previously trained on)
  - Our need statement
  - Our intended use (See lecture slides):
    - Clinicians input patient X-rays into the AI software to diagnose the presence of aseptic loosening in long term orthopaedic implant patients.

# Prototyping Stage 1



# Technical development - prototyping results, research findings, desktop calculations, etc (35%)

- YY
- - previous prototype demonstration video
- - Theory fundamentals -> **dataset preparation** (labelling and fetching) -> prototype execution.
- - HumanSignal sponsorship
- - Stage 2 prototype demonstration

# Prototyping Stage 2

Implant Locator Workstation — B... WhatsApp

127.0.0.1:3000

Implant Locator  
BMEG5552 · UWA

YOLO + resnet58 FastAPI ready Express ready

DEMO PROTOTYPE

STUDY QUEUE

Filter studies

No studies yet. Drop X-ray images below, or anywhere on the window, to build a queue.

Box Heatmap Both Off

Compare 248%

Drop an X-ray anywhere on this window to load a study, then run YOLO detection to place the implant bounding box.

Drop X-ray to add

1 Upload X-ray — 2 Detect + classify — 3 Review box, verdict, heat map

Clear Run analysis

ANALYSIS RESULT

Idle

Select or drop a study, then run the analysis to see the implant bounding box, the loosening verdict, and the Grad-CAM heat map here.

INFERENCE CONTROLS

Reset

Confidence threshold 0.50

Sent to the model on the next run.

IoU (NMS) 0.70

Window / level 248 / 129

W

L

Display only — the model always sees the unmodified pixels.

RUN METADATA

Detector weights/best.pt

Classifier weights/classifier.pt · resnet58

Image size 640 · 320

Classes 3 · Implant, Control, Loose

Latency —

Endpoints /api/predict + /api/classify

Export JSON

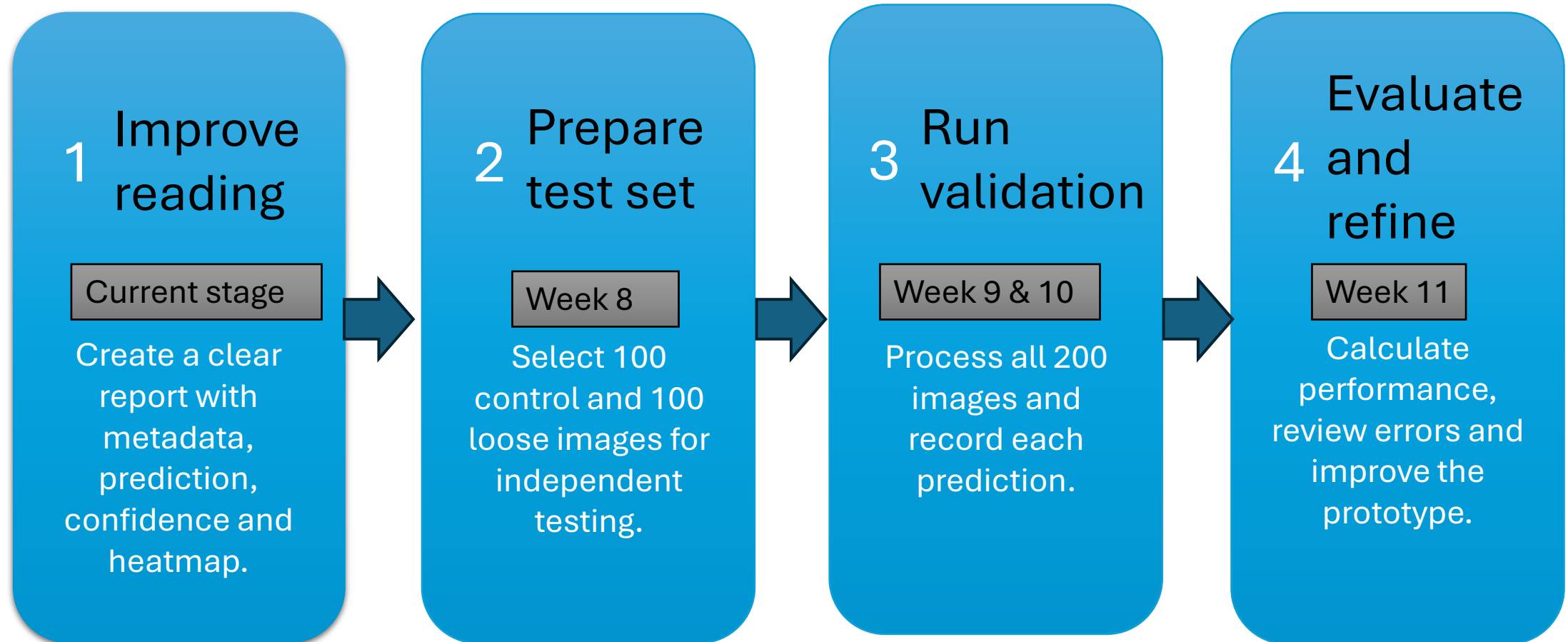
Localisation, loosening classification and Grad-CAM saliency. Trained on 206 images — prototype for educational purposes, not for clinical use.

# Prototyping plan for the rest of the semester (10%)

- **Riah and David Feedback:**

- Need to work out an experiment with potentially 100 control images and 100 loose images to test out the accuracy of our prototype

# Prototyping Plan



# Research and development plan (10%)

- Work on the readability of the results, device should spit out a document that shows all the metadata in a readable way, with the heatmap image that can be exported by the clinician
- Gantt chart (future plans)

Design inputs and risk assessment (35%)